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Tracing Representation Progression: Analyzing and Enhancing Layer-Wise Similarity

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Abstract

Analyzing the similarity of internal representations within and across different models has been an important technique for understanding the behavior of deep neural networks. Most existing methods for analyzing the similarity between representations of high dimensions, such as those based on Centered Kernel Alignment (CKA), rely on statistical properties of the representations for a set of data points. In this paper, we focus on transformer models and study the similarity of representations between the hidden layers of individual transformers. In this context, we show that a simple sample-wise cosine similarity metric is capable of capturing the similarity and aligns with the complicated CKA. Our experimental results on common transformers reveal that representations across layers are positively correlated, with similarity increasing when layers get closer. We provide a theoretical justification for this phenomenon under the geodesic curve assumption for the learned transformer, a property that may approximately hold for residual networks. We then show that an increase in representation similarity implies an increase in predicted probability when directly applying the last-layer classifier to any hidden layer representation. This offers a justification for *saturation events*, where the model's top prediction remains unchanged across subsequent layers, indicating that the shallow layer has already learned the necessary knowledge. We then propose an aligned training method to improve the effectiveness of shallow layer by enhancing the similarity between internal representations, with trained models that enjoy the following properties: (1) more early saturation events, (2) layer-wise accuracies monotonically increase and reveal the minimal depth needed for the given task, (3) when served as multi-exit models, they achieve on-par performance with standard multi-exit architectures which consist of additional classifiers designed for early exiting in shallow layers. To our knowledge, our work is the first to show that one common classifier is sufficient for multi-exit models. We conduct experiments on both vision and NLP tasks to demonstrate the performance of the proposed aligned training.

1 Introduction

As one of the most significant breakthroughs in deep neural network (DNN) architectures developed in recent years, the transformer model [1] has driven recent advances in various vision and NLP tasks, such as vision transformer for image classification [2] and image generation [3–5], BERT [6], GPT [7], and other various large language models (LLMs) [8] for natural language understanding and generation. It has been viewed as a promising foundation model that can be adapted and ex-

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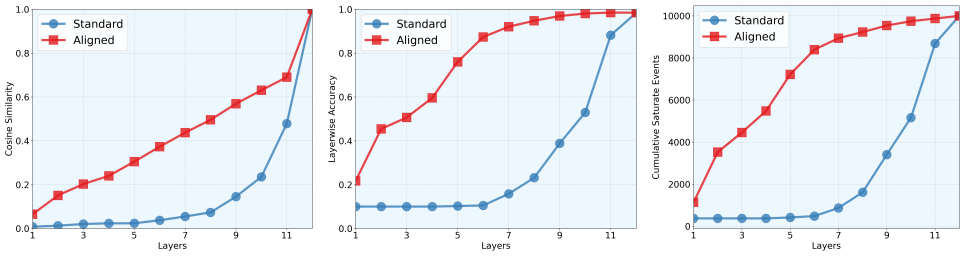

(a) Cosine Similarity (b) Layerwise Accuracy (c) Cumulative Saturation Events

Figure 1: Illustration of the DeiT-S [24] (pretrained on ImageNet) fine-tuned on CIFAR-10 with standard method and the proposed aligned training in terms of (a) cosine similarity of features from shallow and the last-hidden layer, (b) layer-wise testing accuracies by applying the last-layer classifier to each layer, as well as (c) cumulative saturation events [25]. We observe that our proposed aligned training can substantially enhance layer-wise representation similarity, thereby improving layer-wise accuracies and promoting more early saturate events.

tended to various applications and domains [9]. Additionally, researchers have found that increasing the size (by stacking more layers and/or making them wider) can consistently improve performance, resulting in models of significant size (e.g., the 175B-parameter GPT-3 and the 540B-parameter PaLM). However, the ever-increasing size has posed a significant challenge in studying and understanding exactly how these models solve tasks and in efficiently deploying them.

Given the success of deep learning models, attributed to their ability to learn increasingly complex internal representations as they go deeper through their layers, a promising direction for understanding these models is to study the hierarchical feature learning across layers. Recent work [10–14] have uncovered an intriguing phenomenon regarding the last-layer features and classifier of DNNs, called Neural Collapse ($\mathcal{NC}$), across many different datasets and model architectures. Roughly speaking, $\mathcal{NC}$ refers to a training phenomenon in which the last-layer features from the same class become nearly identical, while those from different classes become maximally linearly separable. Beyond the last-layer features, recent studies have also shown that deep classifiers progressively compress within-class features while enhancing the discrimination of between-class features from shallow to deep layers [15–17]. Another line of work attempts to compare the representations within and across DNNs. Various approaches have been proposed to quantify the representation similarity, such as the Canonical Correlation Analysis [18], Centered Kernel Alignment (CKA) [19], Orthogonal Procrustes Transformation [20] and Pointwise Normalized Kernel Alignment (PNKA) [21]. Representation similarity analysis has also been widely used in computational psychology and neuroscience as well [22, 23].

As these approaches are designed to be invariant to certain transformations (such as orthogonal transformation) and can be applied to features with possibly different dimensions, they rely on statistical properties of the representations for entire training data. For instance, the $\mathcal{NC}$ analysis captures the variance of the features from each class, while the widely used CKA for representation analysis replies on the inter-example structures; see Section 2 for the detailed definition of CKA. Consequently, it has been observed that CKA is sensitive to outliers and may give unexpected or counter-intuitive results in certain situations [26]. In this paper, we are motivated by the following question:

How are the representations of individual inputs progressively transformed from shallow to deep layers?

Contribution In this work, we focus on transformer models, which have particular properties compared to other architectures: a transformer contains identical blocks with residual connections in each block. Thus, the features in a transformer model have the same dimension and may exhibit less rotation difference across layers. Motivated by this observation, we study the layer-wise representation similarity for transformer models on a per-sample basis, allowing us to directly apply the last-layer classifier right after any hidden layer for classification or text generation tasks. This enables an effortless multi-exit model that allows early exit during inference, thereby saving computation time. Our contributions can be summarized as follows.

- **Sample-wise cosine similarity captures representational similarity** We introduce a straightforward yet efficient sample-wise cosine similarity metric to examine the similarity of internal representations in transformers. Experiments show that the cosine similarity aligns with CKA, which is based on statistical properties of all the features, and is sufficient to reflect representation similarity. In addition, as illustrated in Figure 1 (a), our experimental results (Standard) on common transformers show that representation similarity increases as layers become closer. We provide a theoretical justification for this phenomenon under the geodesic curve assumption for the learned transformer, a property that approximately holds for residual networks [27], and hence for transformers.
- **Analysis for saturation events** The similarity with last-layer representation suggests that the last-layer classifier can be directly applied right after any hidden layers for decision-making, also known as the logit lens approach. Geva et al. [25] discover a phenomenon called *saturation events*, where the model's final prediction becomes the top candidate in a certain shallow layer and remains unchanged across all subsequent layers, indicating that the shallow layer has already learned the necessary knowledge. We show that an increase in representation similarity implies an increase in predicted probability across layers. This offers a justification for saturation events, stating that if a sample is correctly predicted at the ℓ-th layer, it will continue to be correctly predicted in subsequent layers, as the predicted probability increases progressively across layers.
- **Aligned training for enhancing layer-wise representation similarity** We propose an aligned training method to improve the effectiveness of shallow layers by increasing the similarity of internal representations between different layers. Motivated by the $\mathcal{NC}$ phenomenon, where features from the last-hidden layers align with the common classifier, our aligned training approach deploys the common classifier to each layer and then minimizes the average of the cross-entropy losses from all the layers. As shown in Figure 1, the aligned training can substantially enhance layer-wise representation similarity, thereby leading to more early saturation events and improving layer-wise accuracies. Consequently, the aligned training method can help identify the minimal number of layers needed by unleashing the power of shallow layers to transform features faster towards classifier across layers and push the redundancy behind.
- **Multi-exit models with a single classifier** Another important application of the proposed aligned training is to improve the inference efficiency of large models. During inference, the model allows early exit to save computation time. Previous works [28–30] design multi-exit models by introducing different classifiers to each layer, which may substantially increase the model size for a large number of classes, such as ImageNet with 1000 classes and GPT3 [31] with a vocabulary of $50,257$ tokens. Instead of using separate classifiers for each exit, our multi-exit model employs a common classifier, *which to our knowledge is the first of its kind*, maintaining the early exit capability and achieving performance on par with models that use multiple classifiers. We demonstrate the performance of the proposed aligned training method in both pretraining ViT and fine-tuning LLMs in NLP tasks, including fine-tuning BERT [6] for text classification tasks on the General Language Understanding Evaluation (GLUE) benchmark [32] , and GPT2 [7] model for text generation on the Wikitext-103 dataset [33].

The MatFormer [34] introduces a nested structure into the Transformer by jointly training all submodels of different widths. In contrast, our method jointly trains submodels with varying layers. Exploring the potential integration of these approaches will be a focus of future research. While we mainly focus on transformer models, the proposed aligned training can be applied to other deep architectures, provided they have the same dimensions in each layer. Extending this approach to accommodate varied feature dimensions is the subject of future work.

2 Measuring Layer-wise Representational Similarity

In a transformer f with L layers, the representations gradually evolve across layers, with the progression from one layer (say ℓ-th layer) to the next following a residual update pattern:

$$\boldsymbol{H}^{(\ell+1)} = \boldsymbol{H}^{(\ell)} + f_{\theta^{(\ell)}}(\boldsymbol{H}^{(\ell)}), \tag{1}$$

where $\boldsymbol{H}^{(\ell)} \in \mathbb{R}^{d\times s}$ are input feature sequence of length s with hidden dimension of d. The residual block $f_{\theta^{(\ell)}}(\cdot) : \mathbb{R}^{d\times s} \to \mathbb{R}^{d\times s}$ describe the sequence-to-sequence function mapping with parameters $\theta^{(\ell)}$ that mainly comprises two complementary stages of data transformation: the Multi-head Self-Attention across tokens and the MultiLayer Perceptron layer across features. Predictions are typically based on a specific token of last layer representation $\boldsymbol{H}^{(L)}$. For instance, ViT uses the